

Arhaan Penwala

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EDUCATION

Vellore Institute of Technology

B.Tech Computer Science and Engineering (AI and Robotics), CGPA: **9.13/10.0**

Chennai, India

2023 – 2027

EXPERIENCE

Software Development Engineer Intern

Dec 2025 – Feb 2026

BPO Integra India Private Limited

Remote

- Reduced average page load time by **18%** and improved live-site reliability by designing and shipping 6+ reusable React.js UI components and Node.js/Express.js REST services on a MySQL-backed data layer across 2 sprint cycles, validated through peer code review and unit testing
- Cut average API response time by **22%** across 3 production endpoints by optimizing SQL queries and adding database indexes, collaborating cross-functionally in daily Agile standups to translate requirements into production-ready components

AI Research Intern

May – June 2025

The Indian Hotels Company Limited (IHCL)

Mumbai, India

- Drove enterprise-wide selection of an AI chatbot vendor for a Fortune 500 hospitality group, saving an identified **35%** in projected costs across 5+ vendors and 15+ use cases, by designing and building a structured cost-parametrization evaluation framework
- Directly informed the vendor recommendation adopted by leadership by representing technical requirements in client and vendor demo calls, translating hospitality-specific operational needs into per-use-case cost parameters feeding the evaluation framework

PROJECTS

PolarisGCS | FastAPI, PostgreSQL/TimescaleDB, Redis, React/TypeScript, PySide6, Kotlin, Docker, Kubernetes

4 mos., Ongoing

- Unified command of real and simulated drone fleets under **one MAVLink pipeline** by architecting a 4-component distributed ground control system (React frontend, FastAPI backend, desktop/Android gateways, ArduPilot SITL), containerized with Docker and orchestrated via Kubernetes
- Enabled horizontal scaling of backend instances by synchronizing telemetry via Redis pub/sub and TimescaleDB, then extended platform reach by shipping a Kotlin/Jetpack Compose Android gateway indistinguishable from desktop to the backend

V Help | Flutter, Supabase (PostgreSQL), Firebase, Deno Edge Functions, Azure

1 yr.

- Led development as top contributor (**446/896** commits) over one year on a **178,000-line** Flutter campus super-app spanning food ordering, study materials, hostel/mess, marketplace, and messaging, serving 1,000+ active users
- Secured every privileged operation across the platform by designing **24 serverless** Supabase Edge Functions, and resolved cross-project identity conflicts by bridging two Firebase projects to Supabase auth via a custom JWT-based function

Queez | FastAPI, Flutter, MongoDB, Redis, WebSockets, Gemini API

Ongoing

- Built a real-time multiplayer quiz app (**182/209** commits) that integrates the Gemini API to auto-generate quizzes, flashcards, and notes from uploaded documents
- Eliminated live-session race conditions by implementing per-user/session Redis distributed locks, and hardened scoring integrity against spoofing and overflow exploits with server-side validation

Gait-Based Deepfake Detection | PyTorch, MediaPipe, Grad-CAM, Scikit-learn

2025 – 2026

- Reached **94.95% AUC-ROC** (12.77% EER) on gait-based identity verification by re-architecting a failing Siamese design into a CNN + BiLSTM + Transformer hybrid, validated via 13-fold LOOCV and Grad-CAM interpretability
- Extended research impact by publishing the underlying dataset on IEEE DataPort (DOI: 10.21227/ngh5-b637), with a peer-reviewed journal manuscript currently in progress

TECHNICAL SKILLS

Languages: Python, Java, C/C++, C#, Go, JavaScript/TypeScript, SQL, Kotlin

Frameworks & Libraries: FastAPI, React.js, Node.js, Express.js, Flutter, REST APIs, WebSockets, PyTorch, Scikit-learn, Pandas, XGBoost

Cloud, Containers & Databases: Azure, AWS (EC2, Cognito, S3), Docker, Kubernetes, Firebase, Supabase, Git, PostgreSQL, MySQL, MongoDB, Redis

Systems & Security: Linux (shell scripting), Kali Linux (Bluetooth MITM research), Data Structures & Algorithms, OOP, System Design, Distributed Systems, Agile/Scrum

CERTIFICATIONS

Google AI Essentials, Google Career Certificates

Google Cloud Data Analytics, Google Career Certificates